

Task 3 - Model Card and Governance Stub

Model purpose

Support demonstration of credit default risk scoring, approval decisioning, adverse-action reason generation and fairness monitoring for a jurisdiction-aware RegTech tool.

Intended users

Chief Compliance Officer, Chief Risk Officer, Fair Lending Team, AI Governance Team and Model Risk Management Team. The model is a decision-support artifact, not an autonomous approval authority.

Data

Synthetic credit application data with 5,000 rows. Variables include income, credit score, debt-to-income ratio, loan amount, loan term, employment status, prior default, protected-group proxy, jurisdiction and default outcome. The trained logistic model's predict_proba output is then written as model_score. The data is designed with reference to public lending and credit-risk datasets, but it is not real Citi customer data.

Model and metrics

Primary governance model: logistic regression with standard scaling. Held-out synthetic test performance: AUC 0.800; accuracy 0.876; test observations 1500. Random Forest challenger: AUC 0.793; accuracy 0.785. The challenger tests non-linear signal capture, while the logistic model remains primary because its feature contributions map more directly to adverse-action reasons.

Task 3 - Governance Stub

Jurisdiction logic

Rules are stored in Task3_jurisdiction_rules.csv with jurisdiction, regulatory focus, fairness metric, threshold, threshold basis, explainability requirement, human-review level, documentation level, effective date and rule version. US mode uses a 0.80 internal warning level. EU mode uses Meridian's internal 0.90 governance escalation level, not an AI Act statutory threshold.

Human oversight

The tool flags borderline cases for manual review and escalates portfolio-level fairness issues. Human reviewers must understand the system's limitations, override outputs where appropriate and document remediation decisions.

Error-rate interpretation

The lower protected-group FNR should not be read as simple evidence that the model is fairer to that group. It may indicate that the model is more conservative for protected-group applicants: fewer risky applicants are approved, but more credit access may be denied. The model card therefore reports FNR together with approval parity, FPR and reason distributions.

Failure modes

Rule changes mid-period can make stored parameters stale; mitigation is versioned rules and effective dates. Model drift can reduce accuracy or fairness; mitigation is periodic performance and data drift monitoring. Missing protected attributes can make fairness analysis incomplete; mitigation is explicit limitation flags. Explainability over-reliance can make technical explanations look like legal reasons; mitigation is a clear distinction between model explanation and legal advice.

Task 3 - EU AI Act Obligation Matrix

EU mode is not only a 0.90 threshold. It maps high-risk AI obligations to evidence outputs and owners.

obligation_id	jurisdiction	source	article	control	evidence_output	owner
EU-AIA-10-01	EU	AI Act	Art 10	data suitability and bias checks	data quality report + bias metric	AI Governance
EU-AIA-14-01	EU	AI Act	Art 14	human oversight	reviewer sign-off log	Model Risk
EU-AIA-17-01	EU	AI Act	Art 17	QMS and version control	rule version + validation record	Compliance

Threshold note: the 0.90 threshold is an internal conservative escalation level. Legal duties are represented through the controls above.

Task 3 - EU AI Act Control Mapping

Article 10: data governance

The prototype maps Article 10 to explicit data lineage and quality controls: synthetic-data disclosure, protected-group proxy field, geographic and jurisdiction fields, feature definitions, bias metrics, data-suitability limitations and repeatable sensitivity analysis. In production, this would become source-system lineage, representativeness testing, missing-data controls and bias mitigation records for training, validation and test data.

Article 14: human oversight

The prototype maps Article 14 to reviewer sign-off for borderline applications and fairness escalations. A production workflow would define reviewer competence, override authority, escalation SLAs and records of human decisions.

Article 17: quality management system

The prototype maps Article 17 to rule versioning, effective dates, documented model metrics, validation cadence, human oversight, limitation statements, conflict notes and required governance actions. In production, these would sit inside a quality management system with written policies, change management, post-market monitoring, incident reporting and accountability ownership.

Why the mapping matters

EU mode is not merely a stricter threshold. The 0.90 level is Meridian's internal governance trigger. The AI Act obligations are mapped separately through Article 10 data governance, Article 14 human oversight and Article 17 quality-management evidence.

Task 3 - Operating Controls

Rule update control

Each rule has a version and effective date. If a rule changes mid-period, the tool should preserve prior outputs under the old version, apply the new rule prospectively and require sign-off from Legal, Compliance and Model Risk before changing production thresholds.

Contradictory requirements

If one jurisdiction permits a lower governance threshold while another requires stricter bias testing, the tool does not average the rules. It selects the active jurisdiction for reporting and flags the conflict to a human governance owner. The default product stance is to preserve the stricter evidence trail where cross-border re-use is plausible.

Monitoring cadence

Portfolio fairness metrics should be reviewed monthly, model performance quarterly and rule configuration whenever a material regulatory update occurs. High-risk EU outputs should retain evidence of bias testing, human oversight and documentation review.

Defensible uncertainty

The tool is designed to expose uncertainty. It separates statistical disparity, legal compliance, technical explanation and customer-facing adverse-action reason so management cannot pretend that one metric answers every question.